

CYBE435 HW 03

CYBE435: Computer Incident Response
Homework 3 - Forensics with FTK Imager
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PROMPT

In this assignment, we ask you to do some simple Forensics using the FTK Imager Tool:

1. Install the free version of "FTK Imager" software from the Exterro.com website.
2. On your computer, create a folder named "Lectures" that contains the pdfs of Lectures 1 through 5 of this course, downloaded from the eCampus page.
3. Use FTK Imager to find the MD5 and SHA1 Hash Values for the Lecture #3 pdf. Report these Hash values in the assignment that you submit.
4. Create an AD1 Image of the folder you created using FTK Imager. Verify the image using the built-in hashing feature. Submit the verification log and a screenshot in the assignment that you submit.
5. For your computer, find and report the following items:
 - File system type (NTFS vs FAT)
 - Volume label
 - Number of directories and files
6. Record the MD5 Hash value for the Lecture #4 and Lecture #5 pdfs. Then delete these two files.
7. Use FTK Imager to recover the deleted files from the AD1 image you created in step 4, and place them in a second folder named "Recovered Evidence". Generate MD5 hashes for the recovered files.
8. Select 2 recovered files, and record the following information:
 - Created
 - Modified
 - Accessed
 - Entry Modified (if available)

SUBMISSION

1. Install FTK Imager.

Downloaded from: <https://www.exterro.com/ftk-product-downloads/ftk-imager-pro-8-2-0-26>.

2. Create lecture folder.

```
PS C:\Users\ austi \Documents\cybe435\ftk-imager> ls

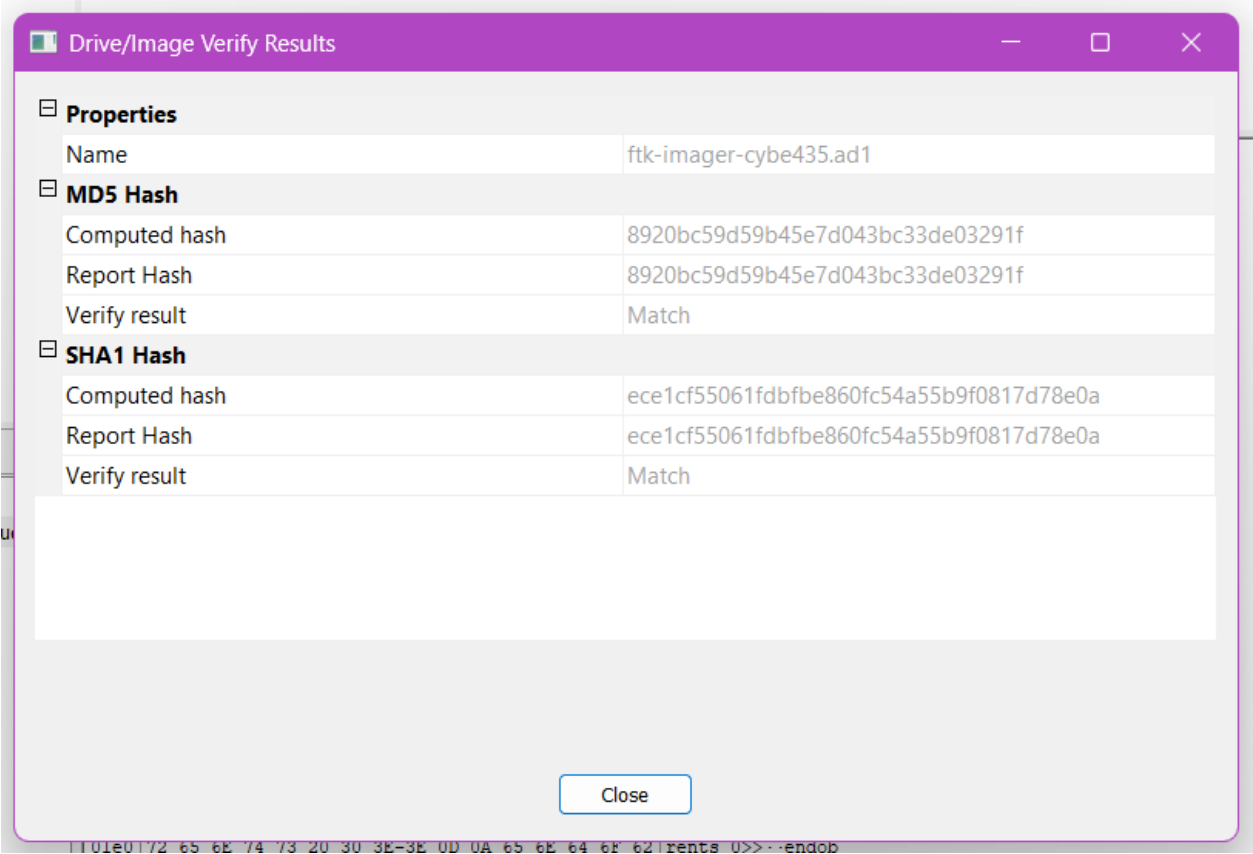
Directory: C:\Users\ austi \Documents\cybe435\ftk-imager

Mode                LastWriteTime         Length Name
----                -
-a- - - -          11/18/2025    7:24 PM      153685 CYBE435 Lecture 01 - Course Overview.pdf
-a- - - -          11/18/2025    7:24 PM      309781 CYBE435 Lecture 02 - Prep and Planning.pdf
-a- - - -          11/18/2025    7:24 PM      18407  CYBE435 Lecture 03 - NCL Intro.pdf
-a- - - -          11/18/2025    7:24 PM      73873  CYBE435 Lecture 04 - Detection and Identification.pdf
-a- - - -          11/18/2025    7:24 PM      99800  CYBE435 Lecture 05 - Detection through Log Analysis.pdf
```

3. Find MD5 and SHA1 Hash.

MD5	SHA1	FileNames
82c53547fc1262ac7e3b26b84411cc82	21f9345e2ba6568b4cf0980e828cb0703e75148e	ftk-imager\C:\Users\ austi\Documents\cyt imager\CYBE435 Lecture 03 - NCL Ir

4. Create an AD1 Image.



MD5 Hash: 8920bc59d59b45e7d043bc33de03291f . SHA1 Hash: ece1cf55061fdbfbe860fc54a55b9f0817d78e0a .

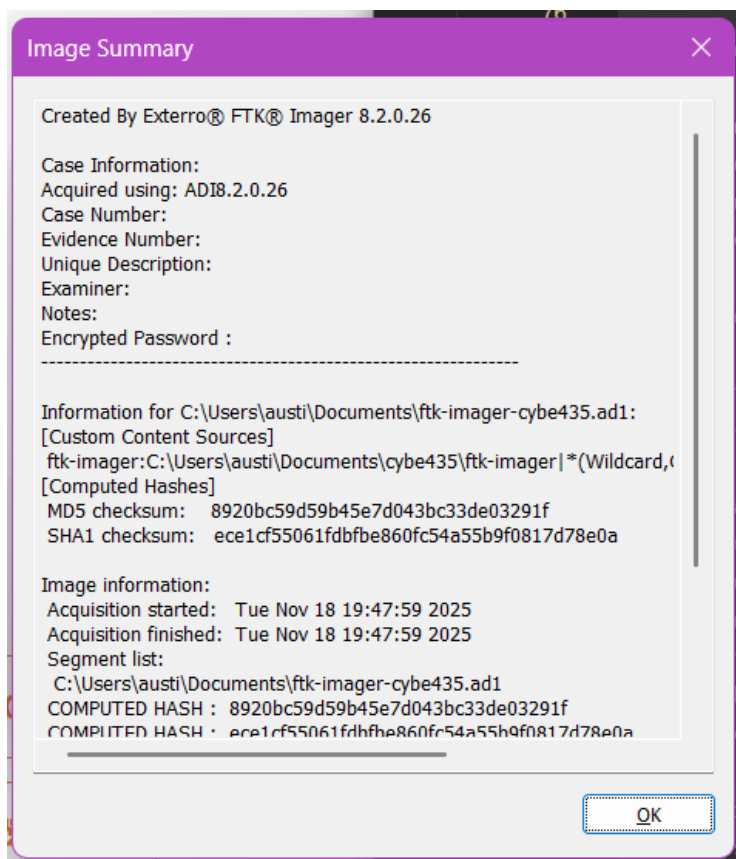


Image Summary:

```
Created By Exterro® FTK® Imager 8.2.0.26

Case Information:
Acquired using: ADI8.2.0.26
Case Number:
Evidence Number:
Unique Description:
Examiner:
Notes:
Encrypted Password :
-----

Information for C:\Users\ austi\Documents\ftk-imager-cybe435.ad1:
[Custom Content Sources]
ftk-imager:C:\Users\ austi\Documents\cybe435\ftk-imager|*(Wildcard,Consider Case,Include Subdirectories)
[Computed Hashes]
MD5 checksum:      8920bc59d59b45e7d043bc33de03291f
SHA1 checksum:     ece1cf55061fdbf860fc54a55b9f0817d78e0a

Image information:
Acquisition started:  Tue Nov 18 19:47:59 2025
Acquisition finished: Tue Nov 18 19:47:59 2025
Segment list:
C:\Users\ austi\Documents\ftk-imager-cybe435.ad1
COMPUTED HASH : 8920bc59d59b45e7d043bc33de03291f
COMPUTED HASH : ece1cf55061fdbf860fc54a55b9f0817d78e0a

Image Verification Results:
Verification started:  Tue Nov 18 19:47:59 2025
Verification finished: Tue Nov 18 19:48:00 2025
MD5 checksum:      8920bc59d59b45e7d043bc33de03291f : verified
SHA1 checksum:     ece1cf55061fdbf860fc54a55b9f0817d78e0a : verified
```

5. Report computer info.

File system type = NTFS, Volume label = Windows, Number of directories = 0, Number of files = 5:

```
PS C:\Users\ austi\Documents\cybe435\ftk-imager> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
-----
C           Windows      NTFS          Fixed    Healthy      OK              60.18 MB      888 MB
C           Windows      NTFS          Fixed    Healthy      OK              98.37 GB     475.79 GB

PS C:\Users\ austi\Documents\cybe435\ftk-imager> (Get-ChildItem "C:\Users\ austi\Documents\cybe435\ftk-imager" -
Recurse -Directory).Count
0

PS C:\Users\ austi\Documents\cybe435\ftk-imager> (Get-ChildItem "C:\Users\ austi\Documents\cybe435\ftk-imager" -
Recurse -File).Count
5
```

6. Record MD5 Hash for Lectures 4 and 5.

Lecture 4: 6b83297840d1ce9c602aa1275673d390 , Lecture 5: 41f7c8e7d75a3464ac3326cfb4553b7f .

Delete the two files:

```
PS C:\Users\ austi\Documents\cybe435\ftk-imager> rm '.\CYBE435 Lecture 04 - Detection and Identification.pdf'
PS C:\Users\ austi\Documents\cybe435\ftk-imager> rm '.\CYBE435 Lecture 05 - Detection through Log Analysis.pdf'
```

7. Generate MD5 hashes for the recovered files.

Recovered Evidence:

```
PS C:\Users\ austi\Documents\cybe435\ftk-imager\Recovered Evidence> ls

Directory: C:\Users\ austi\Documents\cybe435\ftk-imager\Recovered Evidence

Mode                LastWriteTime         Length Name
----                -
-a-----          11/18/2025   7:24 PM           73873 CYBE435 Lecture 04 - Detection and Identification.pdf
-a-----          11/18/2025   7:24 PM           99800 CYBE435 Lecture 05 - Detection through Log Analysis.pdf
```

Lecture 4: 6b83297840d1ce9c602aa1275673d390 . Lecture 5: 41f7c8e7d75a3464ac3326cfb4553b7f .

8. Record recovered file date information.

Recovered File	Date Created	Date Modified	Date Accessed
CYBE435 Lecture 04 - Detection and Identification.pdf	11/19/2025 12:24:54 AM	11/19/2025 12:24:54 AM	11/19/2025 12:24:54 AM
CYBE435 Lecture 05 - Detection through Log Analysis.pdf	11/19/2025 12:24:54 AM	11/19/2025 12:24:54 AM	11/19/2025 12:24:54 AM